

# BLACK RIDER

*An analogue monosynth that sits between the screaming Sallen-Key and the round Moog ladder — two oscillators that drift and lock, a patch bay that wires anything to anything, and the effects a monosynth actually gets played through.*



## BLACK RIDER

two VCO · scream / ladder · patch bay · spring · tape · chorus · drive

## BROKILD

free · VST3 + standalone · 64-bit

Two monosynths define the ground this instrument stands on. One is the Korg MS-20: a Sallen-Key filter with a pair of diodes in its feedback path, which keeps its bass when you turn the resonance up and, past the edge, squeals with harmonics — a thin, eerie, aggressive voice. The other is the Moog: a four-pole transistor ladder whose resonance is round, whose bass thins as the resonance rises, and which sings a clean sine when it takes off — the fat bass every record is built on.

Black Rider is not a copy of either. It is a new instrument that carries both filters and lets you decide, per patch, which circuit the sound goes through. Around them sit two oscillators built to behave like real analogue oscillators — not perfect ones — and a patch bay in the spirit of both the MS-20 and the Moog Grandmother, where the modules stop being wired for you and start being wired by you.

The difference between “nearly there” and “there” is behaviour under pressure. Every stage here does the thing its analogue equivalent does when you push it: the oscillators drift and slowly fall into lock, the mixer and both filters saturate in their own different ways, the envelopes are RC curves and never restart from zero, and each of the three voices has its own component tolerances. None of it is decoration; it is what the ear uses to decide a sound is real.

### What is measured, not hoped

An offline bench renders real audio through the engine and checks the claims by Goertzel analysis and zero-crossing tracking. Both filters self-oscillate within 6 cents of their set cutoff across the range; at 2× and 4× oversampling every alias sits below -60 dB; with VINTAGE at zero the tuning wanders 0.07 cents and the output is dead silent with no note held, while at full VINTAGE it wanders a musical couple of cents. The instrument is warm because those numbers are warm, not because a label says so.

1. Load **Black Rider** as an instrument, or run the standalone. It answers your MIDI keyboard, and it has its own keyboard along the bottom with a pitch and a mod wheel.
2. Press **PATCHES** and pick one. **BLACK BASS** and **ACID RIDER** are the low end; **RIDER LEAD** and **SCREAM SWEEP** show off the Sallen-Key; **THREE RIDERS** and **GHOST ORGAN** are the polyphonic, stereo-spread patches; **FEEDBACK HOWL** is the old MS-20 trick wired through the bay.
3. Turn the two big **CUTOFF** knobs and the **PEAK** beside the low pass. Flip the filter between **SCREAM** and **LADDER** and hear the whole character change under the same setting.
4. Press **PATCH BAY**. The panel grows a wing to the right. Drag a cable from a red hole to a black one

— the LFO into a cutoff, an envelope into a pitch. Roll the mouse wheel over a cable to set how much. Double-click it to pull it out.

5. Press **RANDOM** when you want a whole new machine, and **SAVE** when one is worth keeping.

### The one thing to know

Everything on the panel is one instrument, wired left to right: two oscillators and the extras are summed in the **MIXER**, the mix runs through the **FILTER**, the filter through the **VCA**, and the VCA through the effects to the output. The two envelopes and the LFO shape it; the patch bay lets you break that order wherever you like.



The front panel. Signal flows left to right, top to bottom; the keyboard and wheels are yours to play.

Each oscillator is a phase accumulator with polynomial band-limited steps and ramps, run oversampled, so a saw is a saw all the way up the keyboard without the digital grit of a naïve ramp. Four waves: **saw**, **triangle**, **pulse** (with width and PWM) and **sine**. **RANGE** sets the octave (32' to 4'), **SEMITONE** and **FINE** the interval, **LEVEL** how much reaches the mixer.

What makes them sound like circuits is what they get wrong, on purpose, under **VINTAGE**: a slow temperature walk, a faster jitter, an exponential converter that goes slightly flat at the very top of its range the way a real one does, and per-voice component tolerances so a three-note chord is three slightly different synthesizers. Turn **VINTAGE** down and it is clean and stable; turn it up and it breathes.



VCO 1 — wave, range, semitone, fine, width and PWM.

## VCO 2 — sync, FM and lock

VCO 2 adds three things. **SYNC** hard-syncs it to VCO 1: the reset is band-limited by the actual size of the jump, so a swept sync is a clean, glassy tone, not a bag of aliases. **FM** is exponential modulation of VCO 2 by VCO 1 — small amounts are vibrato and growl, large amounts are bells and clangs. **KBD** lets you take VCO 2 off the keyboard so it holds a fixed pitch, a drone under the notes.

And when the two oscillators sit within a few cents of each other, VCO 2 is weakly pulled towards VCO 1 and the pair slowly falls into lock — exactly as two oscillators on one circuit board do. The **LOCK** lamp on VCO 1 lights when they have. **VINTAGE** sets how strong the pull is.



VCO 2 — the same, plus SYNC, FM and a keyboard switch.



The **MIXER** sums the two oscillators and adds three things of its own: a **SUB** square one or two octaves below VCO 1 (it is divided off VCO 1's own edges, so it carries every wobble VCO 1 has); **NOISE**, white or pink; and **RING**, the ring modulation of the two oscillators. The whole sum passes through **DRIVE**, a saturating stage before the filter — this is where a monosynth gets its body, and turning it up thickens everything downstream.



MIXER — sub, noise, ring, and the pre-filter drive.

### Three filters, one switch

**GROWL** and **SCREAM** are the same two-pole Sallen-Key circuit — a diode clipper in its feedback path, solved as a zero-delay loop — the MS-20 idea. They keep their low end as **PEAK** rises. The difference is how hard **PEAK** pushes: **GROWL** only crosses into self-oscillation near the very top of the dial, so for most of the travel it is a gnarly, thin resonance; **SCREAM** crosses at twelve o'clock, so from noon upward it self-oscillates. Because the oscillation is exactly at the cutoff and the filter tracks the keyboard through **KEY**, a **SCREAM** patch with **KEY** up is a playable sine-ish oscillator in its own right — turn the oscillator **LEVELs** down and play the filter. **LADDER** is a four-pole transistor ladder with saturation in every stage; its resonance is round, its bass thins as **PEAK** rises, and it self-oscillates into a pure sine — the Moog voice. The same patch flipped between the three is three different instruments.

**Measured.** All three self-oscillate within a cent or two of the set cutoff across the range. **SCREAM** sits at 0.08 RMS at noon while **GROWL** is silent there and only wakes near the top; a **SCREAM** tone played an octave apart on the keyboard, **KEY** full, comes out at a 2.00 ratio — in tune.

The filter has a full **HIGH PASS** (with its own **PEAK**) ahead of the **LOW PASS**, so you can hollow the sound out from both ends — a very MS-20 thing to do. Below the cutoffs sit the modulation trims: **EG 1** (how far the filter envelope sweeps the cutoff, either direction), **KEY** (how much the cutoff tracks the note), and **LFO**.



FILTER — high pass, low pass, the SCREAM/LADDER switch, and the modulation trims.

# Envelopes, the LFO and the amplifier

**EG 1** is the filter envelope, **EG 2** the amplifier envelope. Both are the analogue RC shapes — a fast-then-slowing attack, exponential decay and release — and a retrigger continues from wherever the envelope already is, never a click back to zero. **VEL** sets how much key velocity scales each one. The small window in each panel draws the shape and shows the live level.



EG 1 sweeps the cutoff; how far is the EG 1 knob over in the filter.

The **VCA** switch under EG 2 chooses what opens the amplifier: **EG2** (the normal case), **GATE** (a hard on/off, for stabs and organ tones), or **DRONE**

(always open — the synth makes sound with no note held, for the patch bay and for pads).

## The LFO

The **LFO** runs from 0.02 Hz to 2 kHz — slow enough for a minutes-long sweep, fast enough to be an audio-rate modulation source. Six shapes, including S&H and a smooth random DRIFT. **PITCH** is the standing vibrato it always applies; **WHEEL** is the vibrato your mod wheel adds on top; **KEY RESET** restarts its phase on each new note. The LFO is also a source in the patch bay, so it can reach anything.



LFO — six shapes, 0.02 Hz to 2 kHz, vibrato depth and wheel amount.

The **MODE** switch chooses **MONO**, **UNISON** (all three voices on one note, detuned by **DETUNE** and fanned across the stereo field), or **POLY 3** — three-note polyphony. It breaks the monosynth illusion a little, and it is worth it.

**GLIDE** is the portamento; **LEGATO** makes it, and the envelopes, only retrigger when you actually lift off between notes — the classic mono lead behaviour. **TUNE** and **BEND** set master tuning and the pitch-wheel range; **VINTAGE** is the master “how much circuit” control described earlier. **QUALITY** sets the oversampling — 2× is the sensible default, 4× for the most aggressive sync and FM.

**The chord stereo spread.** In **POLY 3**, incoming chords are placed across the stereo field with the **bottom note in the centre** and the higher notes spread left and right — a low root anchored in the middle, the upper voices opened out. Two notes go hard left and right. The **SPREAD** knob sets how wide, from dead centre to fully panned. It is the fastest way to make three oscillators sound like a small string section.



VOICE — mode, spread, detune, glide, tuning, bend, vintage and quality.



The red LED in the header is the **SEED** dial, and it runs from 000 to 199. Every one of those two hundred numbers is a patch: dialling it generates a complete sound — oscillators, filter, envelopes, effects, even patch-bay cables — from the number alone. The same number is always the same instrument, on any machine, in any project. Drag the digits, roll the wheel over them, or use the two arrows; **RANDOM** dials one at random.

The patches are honest about what they are, because each seed is given a **category** first and its parameters are then generated to fit — so the name and the category on the menu always match the sound. The **PATCHES** menu lists all two hundred grouped by category, with your twelve hand-made **SIGNATURE** patches at the top.

| CATEGORY | COUNT | WHAT IT MAKES                                  |
|----------|-------|------------------------------------------------|
| BASS     | 26    | low, mono, EG-swept — the ladder's home ground |
| ACID     | 12    | resonant, gliding, driven — the 303 idea       |
| LEAD     | 24    | mono leads with glide and vibrato              |
| PLUCK    | 20    | fast, short, percussive                        |
| PAD      | 22    | slow, wide, chorused and sprung                |
| STRINGS  | 16    | unison, ensemble-chorused sections             |
| ORGAN    | 16    | drawbar tones, no filter sweep, fast gate      |
| BRASS    | 12    | the filter-swell brass staple                  |
| KEYS     | 14    | bells and electric pianos                      |
| DRONE    | 16    | self-sounding, evolving, no key needed         |
| SYNC     | 10    | hard-synced sweeps                             |
| NOISE    | 12    | percussion and effects                         |

The dial reads the seed; the sound is whatever is on the panel. Dial a seed and everything jumps to it. Tweak a knob afterwards and the sound is yours, though the dial still shows the seed you started from. A saved patch, or a DAW project, keeps the actual values — the seed rides along as a label. A fresh instance opens on seed 000.

Press **PATCH BAY** and the panel grows a wing to the right. Eight cables, each from any **source** (a red hole — the oscillators, the noise, the filter or VCA output, the two envelopes, the LFO, the S&H, and every keyboard control) to any **destination** (a black hole — the pitches, the pulse widths, the cutoffs and peaks, the filter and VCA inputs, the LFO rate, the pan, the drive, the delay time).

Drag from a red hole to a black one to make a cable. Roll the mouse wheel over a cable to set the amount; negative amounts invert and draw as a dashed cord. Double-click a cable, or a patch point, to pull it out. A cable **adds** to whatever the front-panel knob is already doing — it never replaces it.

## Nothing here cares what is “audio” and what is “control”

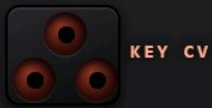
That is the whole point, and it is where the good accidents live. An oscillator into a cutoff is filter FM. An envelope into the delay time is a pitch-shifting echo. The LFO into its own rate is a runaway. And the VCA output back into the filter input is the famous MS-20 feedback patch — wired here in one cable (it is the FEEDBACK HOWL factory patch).

Cables are part of the patch: they save, they load, and they ride in the DAW project with everything else. Each patch point is a block of three holes, so up to three cables can share a source or a destination.

# PATCH BAY

RED HOLES GIVE · BLACK HOLES TAKE  
A CABLE ADDS TO THE KNOB; IT NEVER REPLACES IT

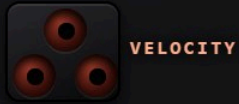
## KEYBOARD



KEY CV



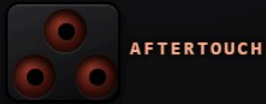
GATE



VELOCITY



MOD WHEEL



AFTERTOUCH



PITCH BEND

## MODULATORS



EG 1



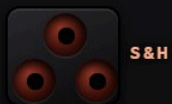
EG 2



LFO



LFO RATE



S&H

## VCO 1



OUT



PITCH



WIDTH



BOTH PITCH

## VCO 2



OUT



PITCH



WIDTH

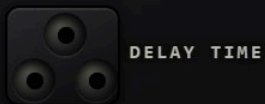


LIN FM

## EFFECTS



DRIVE



DELAY TIME

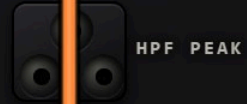
## FILTER



IN



HPF CUTOFF



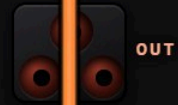
HPF PEAK



LPF CUTOFF



LPF PEAK



OUT

## VCA



IN



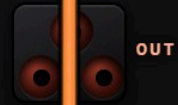
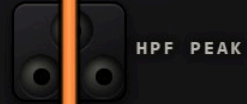
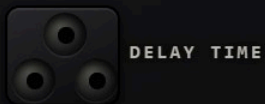
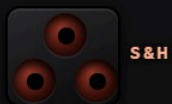
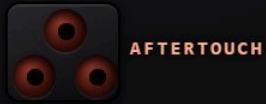
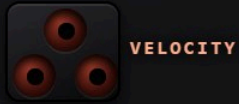
CV

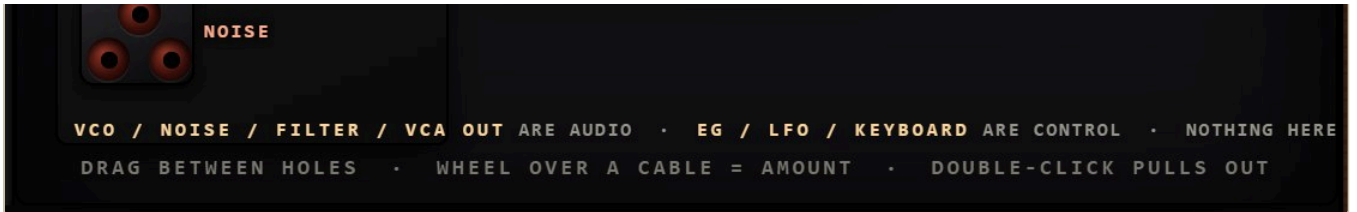


OUT



PAN





The patch bay wing. Red holes give, black holes take; three cables here run EG 1→VCO 2 pitch, LFO→cutoff, and VCA→filter in.

# The effects a monosynth gets played through

DRIVE · CHORUS · TAPE ·  
SPRING

**DRIVE** is an asymmetric overdrive with a biased shaper, so it adds both even and odd harmonics — grit and thickness, with a **TONE** control to keep the top end under control. It runs oversampled, after the voices.



DRIVE.

**CHORUS** is a bucket-brigade (BBD) chorus — a short modulated delay per side with the gentle high-frequency loss and the faint clock whisper of a real BBD chip. **RATE**, **DEPTH**, **MIX**. It is stereo, and it is what turns UNISON into a wall.



CHORUS — BBD.

**TAPE DELAY** is a tape echo, not a clean digital one. **TIME** really does bend the pitch while you move it — the tape has to speed up or slow down to get there. **FEEDBACK**, a **TONE** that darkens each repeat, and **WOW** for the wow-and-flutter of a worn transport. The reels on the panel turn while it runs.



TAPE DELAY — the TIME knob bends the pitch as it moves.

## SPRING is a real spring

The spring reverb is built from three dispersive spring lines, each a long cascade of allpass sections. A real spring passes high frequencies faster than low ones, so every reflection arrives as a downward chirp — that unmistakable “boing”. Here the boing is the dispersion, not an EQ trick. **DWELL** is the decay, **TONE** the brightness, **MIX** the amount.





SPRING — three dispersive springs.

The keyboard along the bottom plays the synth with the mouse (click a key; the vertical position sets velocity) or from your computer keyboard — the home row **Z S X D C V G B H N J M** is an octave, **Q** and **W** shift it down and up. The **PITCH** wheel springs back to centre; the **MOD** wheel stays where you leave it and feeds the LFO's **WHEEL** amount. Both mirror your hardware controller in real time, and both are sources in the patch bay.



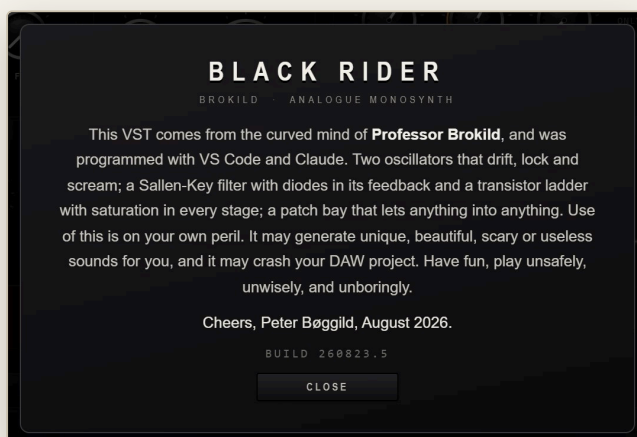
The keyboard, the pitch and mod wheels, and the octave buttons.

At the far right, **OUTPUT** has the master **VOLUME**, a stereo meter with RMS and peak, a **CEILING** lamp that lights when the transparent output limiter is catching a peak, and a **PANIC** button that silences every voice and resets the effects. A soft ceiling sits on the master unconditionally, so nothing here — not a screaming filter, not a feedback patch — can put a spike into your monitors.



OUTPUT — volume, meters, the ceiling lamp and panic.

The **LIGHT** slider in the header lifts the panel's shadows without flattening its contrast, for a brighter room. **ABOUT** tells you whose fault all this is.



ABOUT.

| SOURCE (RED)                        | WHAT IT IS                                                            |
|-------------------------------------|-----------------------------------------------------------------------|
| VCO 1 / VCO 2                       | the raw oscillator, audio — into a cutoff it is filter FM             |
| NOISE                               | white noise, audio                                                    |
| FILTER OUT / VCA OUT                | the signal after the filter, and after the amplifier                  |
| EG 1 / EG 2                         | the filter and amplifier envelopes, 0–1                               |
| LFO / S&H                           | the LFO, and noise sampled once per LFO cycle                         |
| KEY CV / VELOCITY / GATE            | the note as a voltage, how hard it was hit, and whether a key is down |
| MOD WHEEL / AFTERTOUCH / PITCH BEND | the performance controls                                              |
| DESTINATION (BLACK)                 | WHAT IT DOES                                                          |
| VCO1 / VCO2 / BOTH PITCH            | oscillator pitch — an envelope here is a sync or bell sweep           |
| VCO1 / VCO2 WIDTH                   | pulse width                                                           |
| VCO2 LIN FM                         | linear FM of VCO 2 — audio in here makes bells and metal              |
| HPF / LPF CUTOFF, HPF / LPF PEAK    | the filter frequencies and resonances                                 |
| FILTER IN                           | audio straight into the filter — the feedback patch lives here        |
| VCA CV / VCA IN                     | the amplifier's gain, and audio into it past the filter               |
| LFO RATE                            | speeds or slows the LFO, up to four octaves either way                |
| PAN / DRIVE / DELAY TIME            | stereo position, the drive amount, and the tape time (which bends)    |

Everything is automatable from the host except QUALITY (the oversampling), which would glitch if a lane moved it mid-note. Patches save as `.json` files beside the plugin; the DAW project keeps its own full copy.

Black Rider — Brokild, August 2026. Built with VS Code and Claude. This is free software given as-is; play unwisely. The number on the header is the build id — quote it if something is wrong and I will know which one you have.